

Analysis of Linear Programming

Part 3d: Duality

IE 311 – Operations Research I

- Duality is an important concept in optimization.
- We will derive the **dual problem** and learn LP duality.
- We will also discuss the consequences of LP duality and its **economic interpretations**.
- Later, we will learn the matrix form of the simplex method (revised simplex) and the **dual simplex method**, which are closely related to the dual problem.

- Recall that any **Linear Program** can be written as below (equality constraints can be replaced with two inequalities):

$$\begin{aligned}x_0^* &= \max \sum_{j=1}^n c_j x_j \\ \text{s.t. } \sum_{j=1}^n a_{ij} x_j &\leq b_i & i = 1, \dots, m \\ x_j &\geq 0 & j = 1, \dots, n.\end{aligned}$$

- How can we derive a **lower bound** for x_0^* ?
- How can we derive an **upper bound** for x_0^* ?

How to Derive a Lower Bound for a Maximization LP?

- Consider the following LP:
$$\begin{aligned}x_0^* &= \max 2x_1 + 3x_2 \\ \text{s.t. } & x_1 + 2x_2 \leq 5 \\ & x_1 + x_2 \leq 4 \\ & x_1, x_2 \geq 0.\end{aligned}$$
- Observe that
 - $x_0^* \geq 0 = 2 \times 0 + 3 \times 0$ since $(x_1, x_2) = (0, 0)$ is a feasible solution.
 - $x_0^* \geq 5 = 2 \times 1 + 3 \times 1$ since $(x_1, x_2) = (1, 1)$ is a feasible solution.
 - $x_0^* \geq 9 = 2 \times 3 + 3 \times 1$ since $(x_1, x_2) = (3, 1)$ is a feasible solution.
- In general, any **feasible solution** provides a lower bound for a maximization problem and the *best* (largest) lower bound is the optimal value.

How to Derive an Upper Bound for a Maximization LP?

- Consider the following LP: $x_0^* = \max 2x_1 + 3x_2$
s.t. $x_1 + 2x_2 \leq 5$
 $x_1 + x_2 \leq 4$
 $x_1, x_2 \geq 0$.
- Observe that
 - $x_0^* \leq 12$ since $2x_1 + 3x_2 \leq 0 \times (x_1 + 2x_2) + 3 \times (x_1 + x_2) \leq 0 \times 5 + 3 \times 4 = 12$.
 - $x_0^* \leq 10$ since $2x_1 + 3x_2 \leq 2 \times (x_1 + 2x_2) + 0 \times (x_1 + x_2) \leq 2 \times 5 + 0 \times 4 = 10$.
 - $x_0^* \leq 9$ since $2x_1 + 3x_2 \leq 1 \times (x_1 + 2x_2) + 1 \times (x_1 + x_2) \leq 1 \times 5 + 1 \times 4 = 9$.
- In general, any combination of **nonnegative multipliers of the constraints which give an upper bound to the objective function** provides an upper bound.
- What is the set of all such combinations? What is the *best* (smallest) upper bound?

Deriving Bounds for a Maximization LP (continued)

- Consider the following LP:

$$\begin{aligned} x_0^* = \max \quad & \sum_{j=1}^n c_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^n a_{ij} x_j \leq b_i \quad i = 1, \dots, m \end{aligned}$$

- Let us associate a **multiplier** $y_i \geq 0$ for each constraint $i = 1, \dots, m$.
- If we happen to have $\sum_{i=1}^m a_{ij} y_i \geq c_j$ for each variable $j = 1, \dots, n$, then we obtain an upper bound for x_0^* :

- To find the *best* (smallest) upper bound, we should minimize $\sum_{i=1}^m b_i y_i$.

- Consider the *primal* LP:
- The *dual* LP is defined as follows:

$$\begin{aligned} x_0^* = \max \quad & \sum_{j=1}^n c_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^n a_{ij} x_j \leq b_i \quad i = 1, \dots, m \\ & x_j \geq 0 \quad j = 1, \dots, n. \end{aligned}$$

$$\begin{aligned} y_0^* = \min \quad & \sum_{i=1}^m b_i y_i \\ \text{s.t.} \quad & \sum_{i=1}^m a_{ij} y_i \geq c_j \quad j = 1, \dots, n \\ & y_i \geq 0 \quad i = 1, \dots, m. \end{aligned}$$

- Any feasible solution to the dual provides an upper bound to x_0^* !
- Therefore, $x_0^* \leq y_0^*$.

- Recall the primal-dual pair:

$$\begin{array}{ll} x_0^* = \max & \sum_{j=1}^n c_j x_j \\ \text{s.t.} & \sum_{j=1}^n a_{ij} x_j \leq b_i \quad i = 1, \dots, m \\ & x_j \geq 0 \quad j = 1, \dots, n. \end{array} \quad \begin{array}{ll} y_0^* = \min & \sum_{i=1}^m b_i y_i \\ \text{s.t.} & \sum_{i=1}^m a_{ij} y_i \geq c_j \quad j = 1, \dots, n \\ & y_i \geq 0 \quad i = 1, \dots, m. \end{array}$$

- Note that there is a dual (primal) variable for each primal (dual) constraint.
- Hence, the number of dual (primal) variables is equal to the number of primal (dual) constraints.
- The objective coefficients of primal (dual) problem becomes RHS of the dual (primal) problem.
- The RHS of primal (dual) problem becomes the objective coefficients of the dual (primal) problem.

- Recall the primal LP:

$$\begin{aligned}x_0^* = \max \quad & 2x_1 + 3x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 \leq 5 \\ & x_1 + x_2 \leq 4 \\ & x_1, x_2 \geq 0.\end{aligned}$$

- What is x_0^* ?

- Here is the dual LP:

$$\begin{aligned}y_0^* = \min \quad & 5y_1 + 4y_2 \\ \text{s.t.} \quad & y_1 + y_2 \geq 2 \\ & 2y_1 + y_2 \geq 3 \\ & y_1, y_2 \geq 0.\end{aligned}$$

- What is y_0^* ?

Recall the primal LP:

$$\begin{aligned}x_0^* &= \max 2x_1 + 3x_2 \\ \text{s.t. } &x_1 + 2x_2 \leq 5 \\ &x_1 + x_2 \leq 4 \\ &x_1, x_2 \geq 0.\end{aligned}$$

• Notice that $(x_1, x_2) = (3, 1)$ is an optimal solution and $x_0^* = 9$.

Here is the dual LP:

$$\begin{aligned}y_0^* &= \min 5y_1 + 4y_2 \\ \text{s.t. } &y_1 + y_2 \geq 2 \\ &2y_1 + y_2 \geq 3 \\ &y_1, y_2 \geq 0.\end{aligned}$$

• Notice that $(y_1, y_2) = (1, 1)$ is an optimal solution and $y_0^* = 9$.

- Consider the primal LP:
- Here is the dual LP:

$$x_0^* = \max 4x_1 + 3x_2$$

$$\text{s.t. } x_1 + 5x_2 \leq 8$$

$$2x_1 + x_2 \leq 4$$

$$3x_1 + 4x_2 \leq 4$$

$$x_1, x_2 \geq 0.$$

$$y_0^* = \min 8y_1 + 4y_2 + 4y_3$$

$$\text{s.t. } y_1 + 2y_2 + 3y_3 \geq 4$$

$$5y_1 + y_2 + 4y_3 \geq 3$$

$$y_1, y_2, y_3 \geq 0.$$

LP Duality Theorem: Weak Duality

- Consider the following primal-dual pair:

$$\begin{array}{ll} x_0^* = \max & \sum_{j=1}^n c_j x_j \\ \text{s.t.} & \sum_{j=1}^n a_{ij} x_j \leq b_i \quad i = 1, \dots, m \\ & x_j \geq 0 \quad j = 1, \dots, n. \end{array} \quad \begin{array}{ll} y_0^* = \min & \sum_{i=1}^m b_i y_i \\ \text{s.t.} & \sum_{i=1}^m a_{ij} y_i \geq c_j \quad j = 1, \dots, n \\ & y_i \geq 0 \quad i = 1, \dots, m. \end{array}$$

- THEOREM:** We have
the objective value function value of any primal feasible solution $\leq$
the objective value function value of any dual feasible solution.

Hence, $x_0^* \leq y_0^*$.

- Consider the following primal-dual pair:

$$\begin{array}{ll} x_0^* = \max \sum_{j=1}^n c_j x_j & y_0^* = \min \sum_{i=1}^m b_i y_i \\ \text{s.t.} \quad \sum_{j=1}^n a_{ij} x_j \leq b_i & i = 1, \dots, m \quad \text{s.t.} \quad \sum_{i=1}^m a_{ij} y_i \geq c_j \quad j = 1, \dots, n \\ x_j \geq 0 & j = 1, \dots, n. \quad y_i \geq 0 \quad i = 1, \dots, m. \end{array}$$

- THEOREM:** Assume that primal OR dual problem is feasible. Then, optimal value of the primal problem = optimal value of the dual problem, that is, $x_0^* = y_0^*$.
- What would happen if primal is unbounded?
- What if primal is infeasible?

There are three cases for the relation between primal and dual LPs (assume the primal is a maximization problem):

1. The primal is feasible with a **finite** optimal value z_p^* (bounded). Then the dual is feasible and bounded, and $z_p^* = w_d^*$ (strong duality).
2. The primal is feasible and **unbounded** ($z_p^* = +\infty$). Then the dual is **infeasible**.
3. The primal is **infeasible** ($z_p^* = -\infty$). Then the dual is either **unbounded** ($w_d^* = -\infty$) or **infeasible**.

Consider the following primal-dual pair:

$$\begin{aligned}
 x_0^* &= \max \sum_{j=1}^n c_j x_j & y_0^* &= \min \sum_{i=1}^m b_i y_i \\
 \text{s.t. } \sum_{j=1}^n a_{ij} x_j &\leq b_i & i &= 1, \dots, m & \text{s.t. } \sum_{i=1}^m a_{ij} y_i &\geq c_j & j &= 1, \dots, n \\
 x_j &\geq 0 & j &= 1, \dots, n. & y_i &\geq 0 & i &= 1, \dots, m.
 \end{aligned}$$

• **THEOREM:** Let (x^*, y^*) be any pair of primal-dual **optimal** solutions. Then,

$$\left(\sum_{j=1}^n b_i - a_{ij} x_j^* \right) y_i^* = 0, \quad i = 1, \dots, m$$

$$\left(\sum_{i=1}^m a_{ij} y_i^* - c_j \right) x_j^* = 0. \quad j = 1, \dots, n$$

- Consider the primal LP:

$$\begin{aligned}x_0^* &= \max 3x_1 + 4x_2 \\ \text{s.t. } 2x_1 + 3x_2 &\leq 4 \\ x_1 + x_2 &\leq 3 \\ x_1, x_2 &\geq 0.\end{aligned}$$

- What is x_0^* ?

- Here is the dual LP:

$$\begin{aligned}y_0^* &= \min 4y_1 + 3y_2 \\ \text{s.t. } 2y_1 + y_2 &\geq 3 \\ 3y_1 + y_2 &\geq 4 \\ y_1, y_2 &\geq 0.\end{aligned}$$

- What is y_0^* ?
- Complementary slackness?

- Consider the primal LP:

$$\begin{aligned}x_0^* &= \max 3x_1 + 4x_2 \\ \text{s.t. } 2x_1 + 3x_2 &\leq 4 \\ x_1 + x_2 &\leq 3 \\ x_1, x_2 &\geq 0.\end{aligned}$$

- Notice that $(x_1, x_2) = (2, 0)$ is an optimal solution and $x_0^* = 6$.

- Here is the dual LP:

$$\begin{aligned}y_0^* &= \min 4y_1 + 3y_2 \\ \text{s.t. } 2y_1 + y_2 &\geq 3 \\ 3y_1 + y_2 &\geq 4 \\ y_1, y_2 &\geq 0.\end{aligned}$$

- Notice that $(y_1, y_2) = (\frac{3}{2}, 0)$ is an optimal solution and $y_0^* = 6$.

- The primal LP:

$$\begin{aligned}x_0^* &= \max 3x_1 + 4x_2 \\ \text{s.t. } 2x_1 + 3x_2 &\leq 4 \\ x_1 + x_2 &\leq 3 \\ x_1, x_2 &\geq 0.\end{aligned}$$

- The dual LP:

$$\begin{aligned}y_0^* &= \min 4y_1 + 3y_2 \\ \text{s.t. } 2y_1 + y_2 &\geq 3 \\ 3y_1 + y_2 &\geq 4 \\ y_1, y_2 &\geq 0.\end{aligned}$$

- Notice that primal optimal solution $(x_1^*, x_2^*) = (2, 0)$ and dual optimal solution $(y_1^*, y_2^*) = (\frac{3}{2}, 0)$ satisfy **complementary slackness conditions**:

$$(2x_1^* + 3x_2^* - 4)y_1^* = 0$$

$$(x_1^* + x_2^* - 3)y_2^* = 0$$

$$(2y_1^* + y_2^* - 3)x_1^* = 0$$

$$(3y_1^* + y_2^* - 4)x_2^* = 0$$

- Consider the primal LP:

$$\begin{aligned}x_0^* &= \max 3x_1 + 4x_2 \\ \text{s.t. } 2x_1 + 4x_2 &\leq 6 \\ x_1 + x_2 &\leq 5 \\ x_1, x_2 &\geq 0.\end{aligned}$$

- What is x_0^* ?

- Here is the dual LP:

$$\begin{aligned}y_0^* &= \min 6y_1 + 5y_2 \\ \text{s.t. } 2y_1 + y_2 &\geq 3 \\ 4y_1 + y_2 &\geq 4 \\ y_1, y_2 &\geq 0.\end{aligned}$$

- What is y_0^* ?
- Complementary slackness?

- Consider the primal LP:

$$\begin{aligned}x_0^* &= \max 3x_1 + 4x_2 \\ \text{s.t. } 2x_1 + 4x_2 &\leq 6 \\ x_1 + x_2 &\leq 5 \\ x_1, x_2 &\geq 0.\end{aligned}$$

- Notice that $(x_1, x_2) = (3, 0)$ is an optimal solution and $x_0^* = 9$.

- Here is the dual LP:

$$\begin{aligned}y_0^* &= \min 6y_1 + 5y_2 \\ \text{s.t. } 2y_1 + y_2 &\geq 3 \\ 4y_1 + y_2 &\geq 4 \\ y_1, y_2 &\geq 0.\end{aligned}$$

- Notice that $(y_1, y_2) = \left(\frac{3}{2}, 0\right)$ is an optimal solution and $y_0^* = 9$.

- The primal LP:

$$\begin{aligned}x_0^* &= \max 3x_1 + 4x_2 \\ \text{s.t. } 2x_1 + 4x_2 &\leq 6 \\ x_1 + x_2 &\leq 5 \\ x_1, x_2 &\geq 0.\end{aligned}$$

- The dual LP:

$$\begin{aligned}y_0^* &= \min 6y_1 + 5y_2 \\ \text{s.t. } 2y_1 + y_2 &\geq 3 \\ 4y_1 + y_2 &\geq 4 \\ y_1, y_2 &\geq 0.\end{aligned}$$

- Notice that primal optimal solution $(x_1^*, x_2^*) = (3, 0)$ and dual optimal solution $(y_1^*, y_2^*) = (\frac{3}{2}, 0)$ satisfy **complementary slackness conditions**:

$$\begin{aligned}(2x_1^* + 4x_2^* - 6)y_1^* &= 0 \\ (x_1^* + x_2^* - 5)y_2^* &= 0\end{aligned}$$

$$\begin{aligned}(2y_1^* + y_2^* - 3)x_1^* &= 0 \\ (4y_1^* + y_2^* - 4)x_2^* &= 0\end{aligned}$$

- Consider the primal LP:

$$\begin{aligned}x_0^* = \max \quad & 3x_1 + 7x_2 \\ \text{s.t.} \quad & -x_1 + x_2 \leq 6 \\ & x_1 - 2x_2 \leq 5 \\ & x_1, x_2 \geq 0.\end{aligned}$$

- What is x_0^* ?

- Here is the dual LP:

$$\begin{aligned}y_0^* = \min \quad & 6y_1 + 5y_2 \\ \text{s.t.} \quad & -y_1 + y_2 \geq 3 \\ & y_1 - 2y_2 \geq 7 \\ & y_1, y_2 \geq 0.\end{aligned}$$

- What is y_0^* ?

- Consider the primal LP:

$$\begin{aligned}x_0^* = \max \quad & 3x_1 + 7x_2 \\ \text{s.t.} \quad & -x_1 + x_2 \leq 6 \\ & x_1 - 2x_2 \leq 5 \\ & x_1, x_2 \geq 0.\end{aligned}$$

- $x_0^* = +\infty$ since primal is feasible and unbounded.

- Here is the dual LP:

$$\begin{aligned}y_0^* = \min \quad & 6y_1 + 5y_2 \\ \text{s.t.} \quad & -y_1 + y_2 \geq 3 \\ & y_1 - 2y_2 \geq 7 \\ & y_1, y_2 \geq 0.\end{aligned}$$

- Dual is infeasible.

- Consider the primal LP:

$$\begin{aligned}x_0^* = \max \quad & -2x_1 + x_2 \\ \text{s.t.} \quad & 2x_1 - x_2 \leq -2 \\ & -x_1 + x_2 \leq 1 \\ & x_1, x_2 \geq 0.\end{aligned}$$

- What is x_0^* ?

- Here is the dual LP:

$$\begin{aligned}y_0^* = \min \quad & -2y_1 + y_2 \\ \text{s.t.} \quad & 2y_1 - y_2 \geq -2 \\ & -y_1 + y_2 \geq 1 \\ & y_1, y_2 \geq 0.\end{aligned}$$

- What is y_0^* ?

- Consider the primal LP:
- Here is the dual LP:

$$\begin{aligned}
 x_0^* = \max \quad & -2x_1 + x_2 \\
 \text{s.t.} \quad & 2x_1 - x_2 \leq -2 \\
 & -x_1 + x_2 \leq 1 \\
 & x_1, x_2 \geq 0.
 \end{aligned}$$

$$\begin{aligned}
 y_0^* = \min \quad & -2y_1 + y_2 \\
 \text{s.t.} \quad & 2y_1 - y_2 \geq -2 \\
 & -y_1 + y_2 \geq 1 \\
 & y_1, y_2 \geq 0.
 \end{aligned}$$

- Primal is infeasible.
- $y_0^* = -\infty$ since dual is feasible and unbounded.

- Consider the primal LP:

$$\begin{aligned} x_0^* = \max \quad & 2x_1 + x_2 \\ \text{s.t.} \quad & x_1 - x_2 \leq -2 \\ & -x_1 + x_2 \leq 1 \\ & x_1, x_2 \geq 0. \end{aligned}$$

- What is x_0^* ?

- Here is the dual LP:

$$\begin{aligned} y_0^* = \min \quad & -2y_1 + y_2 \\ \text{s.t.} \quad & y_1 - y_2 \geq 2 \\ & -y_1 + y_2 \geq 1 \\ & y_1, y_2 \geq 0. \end{aligned}$$

- What is y_0^* ?

- Consider the primal LP:
- Here is the dual LP:

$$\begin{aligned}
 x_0^* = \max \quad & 2x_1 + x_2 \\
 \text{s.t.} \quad & x_1 - x_2 \leq -2 \\
 & -x_1 + x_2 \leq 1 \\
 & x_1, x_2 \geq 0.
 \end{aligned}$$

$$\begin{aligned}
 y_0^* = \min \quad & -2y_1 + y_2 \\
 \text{s.t.} \quad & y_1 - y_2 \geq 2 \\
 & -y_1 + y_2 \geq 1 \\
 & y_1, y_2 \geq 0.
 \end{aligned}$$

- Primal is infeasible.
- Dual is infeasible.

Dual Problem for Unstructured LPs

[illegible]

- Consider the primal LP:
- Here is the dual LP:

$$\max 3x_1 + 4x_2 - 5x_3$$

$$\text{s.t. } 2x_1 - 7x_2 - 2x_3 \leq 6$$

$$4x_1 + 5x_2 + 4x_3 \geq -2$$

$$3x_1 - 2x_3 = 3$$

$$3x_2 + 4x_3 \geq 3$$

$$x_1 \leq 0, x_2 \text{ urs}, x_3 \geq 0.$$

$$\min 6y_1 - 2y_2 + 3y_3 + 3y_4$$

$$\text{s.t. } 2y_1 + 4y_2 + 3y_3 \leq 3$$

$$-7y_1 + 5y_2 + 3y_4 = 4$$

$$-2y_1 + 4y_2 - 2y_3 + 4y_4 \geq -5$$

$$y_1 \geq 0, y_2 \leq 0, y_3 \text{ urs}, y_4 \leq 0.$$

Dual Problem for Unstructured LPs: Complementary Slackness

$$\begin{aligned}
 \max \quad & 3x_1 + 4x_2 - 5x_3 \\
 \text{s.t.} \quad & 2x_1 - 7x_2 - 2x_3 \leq 6 \\
 & 4x_1 + 5x_2 + 4x_3 \geq -2 \\
 & 3x_1 \quad \quad - 2x_3 = 3 \\
 & \quad \quad 3x_2 + 4x_3 \geq 3 \\
 & x_1 \leq 0, x_2 \text{ urs}, x_3 \geq 0.
 \end{aligned}$$

$$\begin{aligned}
 (2x_1 - 7x_2 - 2x_3 - 6)y_1 &= 0 \\
 (4x_1 + 5x_2 + 4x_3 + 2)y_2 &= 0 \\
 (3x_1 \quad \quad - 2x_3 - 3)y_3 &= 0 \\
 (3x_1 + 4x_2 - 2x_3 - 3)y_4 &= 0
 \end{aligned}$$

$$\begin{aligned}
 \min \quad & 6y_1 - 2y_2 + 3y_3 + 3y_4 \\
 \text{s.t.} \quad & 2y_1 + 4y_2 + 3y_3 \leq 3 \\
 & -7y_1 + 5y_2 \quad \quad + 3y_4 = 4 \\
 & -2y_1 + 4y_2 - 2y_3 + 4y_4 \geq -5 \\
 & y_1 \geq 0, y_2 \leq 0, y_3 \text{ urs}, y_4 \leq 0.
 \end{aligned}$$

$$\begin{aligned}
 (2y_1 + 4y_2 + 3y_3 - 3)x_1 &= 0 \\
 (-7y_1 + 5y_2 \quad \quad + 3y_4 - 4)x_2 &= 0 \\
 (-2y_1 + 4y_2 - 2y_3 + 4y_4 + 5)x_3 &= 0.
 \end{aligned}$$

Dual Problem for Unstructured LPs (continued)

$$[\mathbf{P2}] \max 2x_1 + x_2 + 3x_3$$

$$\text{s.t. } x_1 + x_2 + x_3 = 3$$

$$x_1 - 2x_2 + x_3 \geq 1$$

$$2x_2 + x_3 \leq 2$$

$$x_1 \geq 0, x_2 \leq 0, x_3 \text{ urs.}$$

$$[\mathbf{D2}] \min 3y_1 + y_2 + 2y_3$$

$$\text{s.t. } y_1 + y_2 \geq 2$$

$$y_1 - 2y_2 + 2y_3 \leq 1$$

$$y_1 + y_2 + y_3 = 3$$

$$y_1 \text{ urs.}, y_2 \leq 0, y_3 \geq 0$$

Solution & Obj. Func. Val.	Solution & Obj. Func. Val.

Consider the following problem.

$$\begin{array}{ll}\min & c_2 x_2 - x_1 - x_3 \\ \text{s.t.} & x_3 - x_1 - x_2 \leq 1 \\ & 2x_1 - x_2 + x_3 \leq 2 \\ & -2 \leq x_1 \leq 1 \\ & -2 \leq x_3 \leq -1\end{array}$$

- (a) Construct the dual problem.
- (b) Use duality theory and explain if $(x_1, x_2, x_3) = (-1, 2, -1)$ can be an optimal solution. Does your answer depend on c_2 ? If so, provide necessary conditions.
- (c) Use duality theory and explain if $(x_1, x_2, x_3) = (1/3, -10/3, -2)$ can be an optimal solution. Does your answer depend on c_2 ? If so, provide necessary conditions.
- (d) Provide a feasible corner point solution (x_1, x_2, x_3) other than those provided above and obtain the corresponding dual solution

- Consider the following primal-dual pair again:

$$x_0^* = \max \sum_{j=1}^n c_j x_j$$

$$\text{s.t.} \quad \sum_{j=1}^n a_{ij} x_j \leq b_i \quad i = 1, \dots, m$$

$$y_0^* = \min \sum_{i=1}^m b_i y_i$$

$$\text{s.t.} \quad \sum_{i=1}^m a_{ij} y_i \geq c_j \quad j = 1, \dots, n$$

- Let us think of the primal variables $x_j \geq 0$ as the amount of different items produced and primal constraints as common resource limitations.
- Dual variables are sometimes called **shadow prices** as they give an idea about the price of marginal increase in the respective resource.

- Giapetto's, Inc., manufactures two types of wooden toys: soldiers and trains.
- Each soldier built:
 - sells for \$27 and uses \$10 worth of raw materials.
 - increases Giapetto's variable labor/overhead costs by \$14.
 - requires 2 hours of finishing labor and 1 hour of carpentry labor.
- Each train built:
 - sells for \$21 and used \$9 worth of raw materials.
 - increases Giapetto's variable labor/overhead costs by \$10.
 - requires 1 hour of finishing labor and 1 hour of carpentry labor.
- Each week Giapetto has only 100 finishing hours and 80 carpentry hours.
- Giapetto wants to maximize weekly profit (revenues - costs).
- See **GiapettoDual.py** to see how to retrieve optimal primal and dual solutions.
- WATCH: www.loom.com/share/db92154bc8e54e8b96fc6b039e35c812

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- Each week Giapetto has only 100 finishing hours and 80 carpentry hours.
- Giapetto wants to maximize weekly profit (revenues - costs).
- See **GiapettoDual.py** to see how to retrieve optimal primal and dual solutions.

- x_1 : number of soldiers

- x_2 : number of trains

- The primal LP:

$$\begin{aligned}x_0^* &= \max 3x_1 + 2x_2 \\ \text{s.t. } 2x_1 + x_2 &\leq 100 \\ x_1 + x_2 &\leq 80 \\ x_1, x_2 &\geq 0.\end{aligned}$$

- Here is the dual LP:

$$\begin{aligned}y_0^* &= \min 100y_1 + 80y_2 \\ \text{s.t. } 2y_1 + y_2 &\geq 3 \\ y_1 + y_2 &\geq 2 \\ y_1, y_2 &\geq 0.\end{aligned}$$

- $(x_0^*, x_1^*, x_2^*) = (180, 20, 60)$

- $(y_0^*, y_1^*, y_2^*) = (180, 1, 1)$

- Would you purchase an additional unit of Resource 1 for \$2?
- The modified primal LP with $(x_0^*, x_1^*, x_2^*) = (181, 21, 59)$:
$$x_0^* = \max 3x_1 + 2x_2$$
$$\text{s.t. } 2x_1 + x_2 \leq 101$$
$$x_1 + x_2 \leq 80$$
$$x_1, x_2 \geq 0.$$
- Observe that the additional benefit of 1 unit of Resource 1 is only \$1, therefore, we would not purchase it.
- Recall that the dual variable (shadow price) of Resource 1 was \$1. Therefore, we had not had to solve the modified LP!

- Would you purchase an additional unit of Resource 1 for \$0.5?
- Recall that the modified primal LP is with $(x_0^*, x_1^*, x_2^*) = (181, 21, 59)$.
- Since the additional benefit of 1 unit of Resource 1 is \$1, we would purchase it now.
- Recall that the dual variable (shadow price) of Resource 1 was \$1.

- Would you purchase 10 units of Resource 1 for \$0.5 each?
- Recall that the modified primal LP is with $(x_0^*, x_1^*, x_2^*) = (190, 30, 50)$.
- Since the additional benefit of 10 unit of Resource 1 is \$10, we would purchase it.
- Recall that the dual variable (shadow price) of Resource 1 was \$1 and we can *predict* the increase in the profit as 1×10 .

- Note that the benefit of additional unit of resources might diminish after a certain point.
- For instance, if the RHS of Resource 1 is increased from 100 to 160, then the additional profit is \$60 (as predicted by 1×60).
- However, if the RHS of Resource 1 is increased from 100 to 161, then the additional profit is still \$60.
 - Later, we will see how to compute this *critical point* 60.
- Therefore, the actual increase in the profit might be less than the prediction of **shadow price $\times$ additional units purchased**.

- Consider the following primal-dual LP pair.

$$\begin{array}{ll} x_0^* = \max \sum_{j=1}^n c_j x_j & y_0^* = \min \sum_{i=1}^m b_i y_i \\ \text{s.t.} \quad \sum_{j=1}^n a_{ij} x_j \leq b_i & \text{s.t.} \quad \sum_{i=1}^m a_{ij} y_i \geq c_j \quad j = 1, \dots, n \\ x_j \geq 0 & y_i \geq 0 \quad i = 1, \dots, m. \end{array}$$

- Recall that we interpreted optimal dual variable y_i^* as the **shadow price** of resource i (maximum amount to pay to acquire an additional unit of resource).
- We also **predicted** the change in the optimal value as $\Delta \times y_i^*$ if resource i is increased by Δ .
 - In fact, this is an upper bound on the change. Let us now understand why.

- Consider the following primal-dual LP pair.

$$\begin{array}{ll} x_0^* = \max & \sum_{j=1}^n c_j x_j \\ \text{s.t.} & \sum_{j=1}^n a_{ij} x_j \leq b_i \quad i = 1, \dots, m \\ & x_j \geq 0 \quad j = 1, \dots, n. \end{array} \quad \begin{array}{ll} y_0^* = \min & \sum_{i=1}^m b_i y_i \\ \text{s.t.} & \sum_{i=1}^m a_{ij} y_i \geq c_j \quad j = 1, \dots, n \\ & y_i \geq 0 \quad i = 1, \dots, m. \end{array}$$

- Let (x^*, y^*) be a pair of primal-dual optimal solutions.
- Suppose that the RHS of primal constraint i is increased by Δ .
 - Note that the dual feasible region is unchanged!
 - Therefore, y^* is still a feasible solution and its objective value changes by $\Delta \times y_i^*$.
 - Since y^* may not be optimal for the new dual problem anymore, we obtain an **upper bound** on the change. Finally, we use strong duality.

- Consider the following primal-dual LP pair.

$$\begin{array}{ll} x_0^* = \max & \sum_{j=1}^n c_j x_j \\ \text{s.t.} & \sum_{j=1}^n a_{ij} x_j \leq b_i \quad i = 1, \dots, m \\ & x_j \geq 0 \quad j = 1, \dots, n. \end{array} \quad \begin{array}{ll} y_0^* = \min & \sum_{i=1}^m b_i y_i \\ \text{s.t.} & \sum_{i=1}^m a_{ij} y_i \geq c_j \quad j = 1, \dots, n \\ & y_i \geq 0 \quad i = 1, \dots, m. \end{array}$$

- Let (x^*, y^*) be a pair of primal-dual optimal solutions.
- Suppose that the objective coefficient of primal variable j is increased by Δ .
 - Note that the primal feasible region is unchanged!
 - Therefore, x^* is still a feasible solution and its objective value changes by $\Delta \times x_j^*$.
 - Since x^* may not be optimal for the new primal problem anymore, we obtain a **lower bound** on the change.

• Primal LP:

$$\begin{aligned}x_0^* &= \max 3x_1 + 2x_2 \\ \text{s.t. } 2x_1 + x_2 &\leq 100 \\ x_1 + x_2 &\leq 80 \\ x_1, x_2 &\geq 0.\end{aligned}$$

• Dual LP:

$$\begin{aligned}y_0^* &= \min 100y_1 + 80y_2 \\ \text{s.t. } 2y_1 + y_2 &\geq 3 \\ y_1 + y_2 &\geq 2 \\ y_1, y_2 &\geq 0.\end{aligned}$$

- $(x_0^*, x_1^*, x_2^*) = (180, 20, 60)$
- $(y_0^*, y_1^*, y_2^*) = (180, 1, 1)$
- Predict the change in the primal and dual optimal value if
 - the objective coefficient of x_1 increases by 1. How about 2?
 - the objective coefficient of x_2 decreases by 1. How about 2?

- The Dakota Furniture Company manufactures desks, tables, and chairs.
- The manufacture of each type of furniture requires lumber, finishing and carpentry.
- The amount of each resource needed to make each type of furniture, availability of each resource and selling price of each furniture are given as below:

	Desk	Table	Chair	Availability
Lumber (board ft)	8	6.5	1.5	48
Finishing hrs	4	2.5	1.5	20
Carpentry hrs	2	1.5	0.5	8
Price (\$)	60	30	20	

- Formulate and solve an LP to find the optimal primal and dual solutions using Python-Gurobi (see **DakotaFurniture_Template.py** uploaded to Sucourse).
- Would you purchase an additional unit of
 - Lumber for \$1? Finishing hour for \$5? Carpentry for \$15?

Predict the change in the primal and dual optimal values if:

- the objective coefficient of x_1 increases by 10;
- the objective coefficient of x_2 increases by 5;
- the objective coefficient of x_3 decreases by 4.